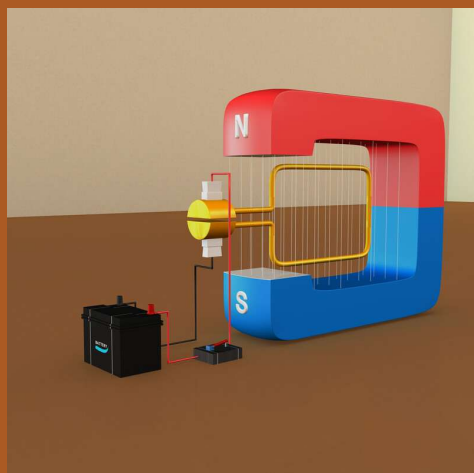


Worksheet 2

Understanding torque and mass

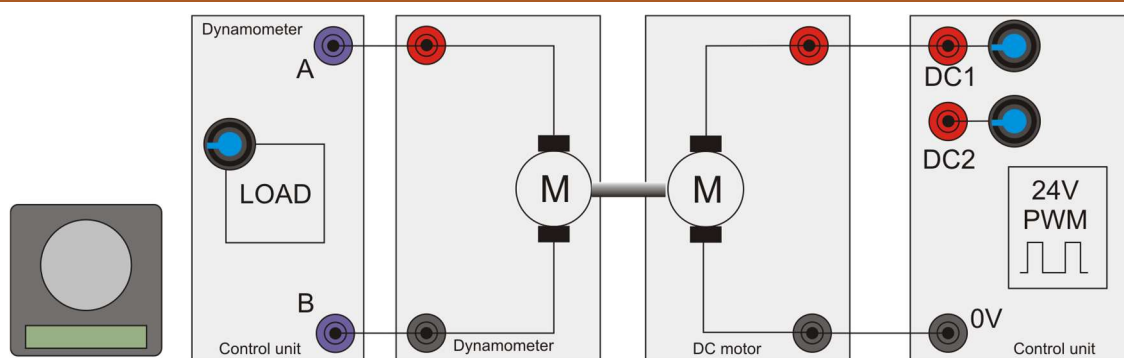
Modern electrical machines



DC motors use a permanent magnet in the stator and direct current powered windings in the rotor.

A commutator is used to reverse the direction of the current in the rotor to maintain the force on it as the motor turns.

The graphic shows a very simple commutator. In practice, the ring has several splits in it, allowing several rotor circuits to provide more continuous torque.



Over to you:

With the control box in manual mode:

- 1) Put the balance under the swing arm and turn it on.
- 2) Turn the control unit potentiometers to minimum (fully anti-clockwise.)
- 3) Slowly increase the value of DC1 until the DC motor starts to turn.
- 4) Is there a reading on the balance?
If not, swap over the positive and negative connections on the motor to change the direction of rotation.
- 5) With DC1 potentiometer in the middle of its range, note of the balance reading and motor speed in the table below. (The motor speed will vary a little - estimate an average.)
- 6) Turn the DC1 PSU pot to increase the voltage slightly. Measure and record the new balance

reading and speed.

- 7) Referring to the 'Key Relationships' (page 4,) use the readings to calculate the torque for the two settings and record your answers .

Speed	Balance torque

So what?

To change the direction of a DC motor you reverse the electrical connections.

The swinging-arm dynamometer can rotate in either direction. The torque produced is the same for both. When rotating counter-clockwise, pressure is applied to the balance. When rotating clockwise, the pressure is on the load cell at the back of the dynamometer.